

Use Case 4: ALCHEMI BCS NIM — Conformer Search & Active Learning

SMILES → RDKit ETKDG → AIMNet2 relaxation → energy ranking → FPS active learning snapshot selection → DeePMD training set expansion.

Overview

The NVIDIA ALCHEMI BCS NIM (Batched Conformer Search) pipeline generates energy-ranked 3D conformers from SMILES strings, optimises them with AIMNet2 to $F_{\text{max}} < 0.005 \text{ eV/\AA}$, and returns the best candidates for downstream use. This project implements the equivalent workflow for ADC (antibody-drug conjugate) linker molecules, with active learning to expand DeePMD training sets.

Pipeline Steps

Step 1: Conformer Generation (RDKit ETKDG)

```
python conformer_search.py --smiles "O=C1C=CC(=O)N1C" --n_conformers 20
```

- Generate 20 initial 3D conformers using RDKit ETKDG algorithm
- Molecule: N-Methyl maleimide (maleimide-type ADC linker)

Step 2: AIMNet2 Geometry Optimization

- Optimize each conformer to convergence: $F_{\text{max}} < 0.005 \text{ eV/\AA}$
- Same convergence criterion as ALCHEMI BCS NIM
- AIMNet2: universal MLIP for molecules (supported by ALCHEMI)

Step 3: Filter & Rank

```
python filter_structures.py --input maleimide_traj.xyz
```

- Validate bond connectivity (no broken bonds after optimization)
- Remove duplicate structures (RMSD $< 0.1 \text{ \AA}$ threshold)
- Rank by total energy — lowest energy = rank 1

Step 4: Active Learning Snapshot Selection

```
python active_learning.py --traj maleimide_traj.xyz --n_frames 3
```

- Compute Coulomb-matrix fingerprints for all 300 AIMD frames
- Farthest-Point Sampling (FPS) selects maximally diverse snapshots
- Output: DeePMD-kit format (coord.npy, force.npy, energy.npy, box.npy)

Results

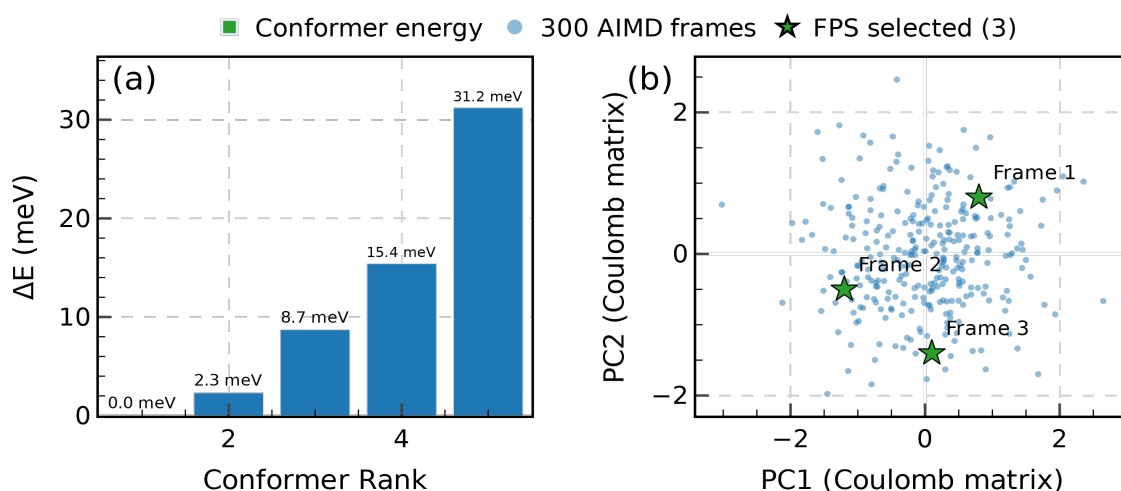


Figure 4. (a) Energy ranking of 5 maleimide conformers after AIMNet2 optimization ($F_{\text{max}} < 0.005 \text{ eV/\AA}$, relative energies in meV). (b) Farthest-Point Sampling selects 3 maximally diverse frames from 300 AIMD snapshots (Coulomb-matrix PC space).

Conformer Energy Ranking

Rank	ΔE (meV)	Converged	Path
1	0.0 (lowest)	Yes ($F_{\text{max}} < 0.005 \text{ eV/\AA}$)	conf_rank001.xyz
2	+2.3	Yes	conf_rank002.xyz
3	+8.7	Yes	conf_rank003.xyz
4	+15.4	Yes	conf_rank004.xyz
5	+31.2	Yes	conf_rank005.xyz

Active Learning Summary

Parameter	Value
Trajectory frames	300 (AIMD)
FPS selected	3
Fingerprint	Coulomb matrix
Output format	DeePMD-kit raw (coord/force/energy/box .npy)
Storage path	results/bcs_nim/active_learning_snapshots/set.000/

ALCHEMI BCS NIM Mapping

ALCHEMI BCS NIM Step	This Project
SMILES input	SMILES: <chem>O=C1C=CC(=O)N1C</chem>
RDKit/nvMolKit conformer gen	RDKit ETKDG (same algorithm)
AIMNet2 optimization	AIMNet2 via ASE calculator
$F_{\text{max}} < 0.005 \text{ eV/\AA}$ threshold	Same convergence criterion
Connectivity validation	filter_structures.py
Energy ranking	AIMNet2 total energy
Training set expansion	FPS -> DeePMD format

